IMMC 2023 International Round—Problem F

Team Control Number: IMMC23255691

I. Abstract

The level of profit and benefits that a piece of land gets undoubtedly depends largely on the planning of the area, and effective planning can set the road to success in motion, building the most fundamental base for ensuring the prosperity in the region. Therefore, in this time's question, we are assigned a parcel of land to determine the best metric of apportioning it for the most beneficial constructions among the given options so that it bears the most fruit.

During the initial planning, our team identified 8 factors that will affect the total benefit of the project: cost upfront; cost sustained; earn; land; environment; jobs; attraction; tourist. Cost upfront describes the amount of money that is paid before the construction is finished, and cost sustained describes the costs that are incurred as a result of sustaining activities, including an allowance for funds used during the period of construction.

Our team connected these factors and grouped them into 3 main categories based on their characteristics: economical, environmental, and human. This allowed us to view the problem in a more objective way as we now had the means to target specific pieces of data for comparison between the values.

Our economy section contains calculations of GDP and profit, as well as upfront and sustained costs. We utilized GDP and profit as the main components to calculate our economic factor. Through adding some coefficients, we were able to scale the numbers down so that the values for economy were aligned on the same scale as the others, preventing it from becoming the dominant factor in our overall model. Environment section contains land and environment. We identified depletion in soil, and changes in terrain as the two main factors. Adding some coefficients, we made sure that the factor will not be dominant. Human section contains jobs, attraction, and tourism. We then added all of these factors together, scaling down through the ratio of tourist's maximum and the jobs' maximum. We also added some coefficients to scale it down furthermore. Finally, the benefit is around 0 – 100, with an average of about 50. With research, we found what we wanted, and inputted the data to come to the final benefits.

Moving on, we saw that the third question required us to analyze the effects of a nearby factory will have upon the construction and facilities on our assigned land. After consideration our team identified three factors that would bring about : electricity cost, population increase, number of jobs. These will affect some numbers of our model. Last but not least, question four requests a generalization model based on the places we are familiar with. We summarized three main differences between the great metropolis we are familiar with—Shanghai— and the piece of land the problem provided. There would be a larger population, a worse atmospheric environment and a more developed economy. Therefore, we would have to change some inputted values and constants into variables to generalize our model.

Overall, in order to successfully produce the model for the given problem, we conducted extensive research to identify the most significant factors, finally producing the concise and widely applicable model that would be explained in detail in the following pages.

Table of Contents

II. Introduction

Efficient land use planning is undoubtedly important when facing a piece of undeveloped land. It is key to paving the road for transforming the land into a profitable one, providing a sturdy base on which later actions can take place. Therefore, laying down a solid and foolproof plan for land usage is a fundamental basis to the profiting and success of the land owners. Yet, planning comes along with the need to take a wide variety of different factors into account, such as unique geography, network of roads and transportation, climate, etc., so a wide perception of the problem would be a definite need to allow for the thriving for businesses and commercial centers in the said area of land. And this is not even where the benefits of an efficient land use plan stop. By identifying the connection and differences between different regions, it may even serve as a further blueprint for the planning of other areas of land, applicable to a wider range of real-life terrains while also allowing each one to be utilized to its specific maximum limit.

This is what our team strives towards as we are assigned to devise the best plan for an area of land located in Syracuse, NY, USA. Therefore, by identifying the main factors that we will focus on and combining data we have done research on, we will produce a model which will be able to let the land go to its full extent and even potentially pose as a reference frame for future reconstructions.

However, before explaining our model, we will first begin by restating the question.

III. Defining the problem

3.1 Question restatement

In this time's problem, we are assigned to decide on the best use of a parcel of land with an area of 3 squared kilometers in Syracuse, NY, America, and which has a temperate marine climate including a snowy winter and adequate water and power supplies. With those conditions in mind, we are asked to complete the following tasks:

1. Determine a quantitative decision metric that defines "best" and considers both short and long term benefits and costs.
2. Choose at least 2 of the choices: outdoor sports complex, cross-country skiing facility (3-month season), a crop farm, a grazing farm/ranch, a regenerative farm, a solar array, an agrivoltaic farm, and an agritourist center. Then decide on the factors to focus on and why.
3. Predict how the new fab announced to be built in Clay, NY, USA, would impact our metric and reevaluate the options we have made previously.
4. Briefly discuss how well our model could be implemented in an environment we are familiar with as well as consider the generalizability our model is to other locations.
5.

Lastly, we are to compose a one-page letter communicating the key details of our recommendation to the decision-makers.

3.2 Assumptions and Justifications

Assumptions	Justifications
We assume that the government has supreme control over the area, such that no additional costs are required to move occupants out of the region of land.	If some people are unwilling to leave the place and go against the reconstruction, our model would become too inaccurate since there are numerous unpredictable factors which we would not be able to take under account. Besides, the cost required to drive people out of the region would be low compared to the cost of establishing new architecture, so this assumption will not affect our model in any significant way.
We assume all existing trees in the region are the common pine trees species, costing \$30 for each	We take to consideration the price of trees because selling them could help increase the GDP of our model. Research about tree species showed that different trees differed greatly in cost, so we only considered pine trees to make the model more comprehendible. Moreover, the climate of Red Creek is cold and humid, proposing that the most common type of tree would be pine. Further research also showed the average cost per pine tree to be \$30.
We assume that the section of land only contains forests, crops and wetlands.	Considering more types of landscapes would result in a more complex model with more variables taken into consideration, and it is unreasonable for 3 squared kilometers to bear too much terrains. Since these areas cover up almost 100% of the land, we will simplify our model by only including these three landscapes.
We assume that the elevation of the area does not affect the efficiency of our model such as cost for construction or growth of crops.	According to the terrain data provided by Caltopo, the maximum difference in elevation in the 3.1km^2 area is only 26 meters. Since the area is relatively large and mostly flat, the difference in elevation would be insignificant.
We assume that there will be no large events or fluctuations that could affect the economics or population of Red Creek.	Large events such as the COVID-19 pandemic or the Great Depression are both hard to predict and scarce, and so as a result we do not need to take them into

	consideration. Taking too many factors into consideration would render a complicating model that may be inaccurate.
We assume that during the entirety of both the construction and the actual opening of the reconstruction there will be no significant natural disaster or accidents.	The area that Red Creek is situated in (NY state), is not prone to experiencing costly disasters, and if there were major accidents it would be impossible to predict in our model the exact parts, so making this assumption prevents unnecessary complication of the problem.

3.3 Variables

Variable	**Definition**
M_{cu}	Cost upfront - the amount of money required to reconstruct the land based on our quantitative decision
M_e	Earn - the annual income of a specific way of using the provided land
M_{cs}	Cost sustained - the annual payment required for maintenance of the structures within the land
l_f	The area of the original piece of land containing forests
l_c	The area of the original piece of land suitable for growing crops (mostly plains)
l_w	The area of the original piece of land which has high water composition in its soil
λ	The set of the three variables l_f, l_c and l_w where each variable has an assigned value
l_{fa}	The area of the piece of land containing forests after using the land
l_{ca}	The area of land suitable for growing crops after using the land
l_{wa}	The area of land which still has high water composition in its soil after using the land

λ_a	The set of the three variables l_{fa}, l_{ca} and l_{wa} where each variable has an assigned value
dp	The percentage of the amount of nutrients in the soil of the land decreased within one year
A_e	The number of employees a way of using the land requires within one year
A_{mw}	The number of migrant workers the project will attract within one year
A_t	The number of tourists that will be attracted to this parcel of land within one year

IV. About the Algorithm

4.1 Overview

The major factors we have chosen to take into consideration are: GDP, jobs provided, and the environmental impact of our proposal. In our model we have decided to weigh the importance of these three factors depending on their relative importance to the region.

We believe the most emphasis should be placed on the factor of available jobs to increase employment rates and decrease poverty in the region. According to statistical data from the ACS 5-year estimates (US Census Bureau), the proportion of those suffering from poverty in the Red Creek Central School District is approximately 1.3 times that of the surrounding Cayuga County and New York State. If jobs with stronger expertise in certain regions could be created, it would also serve as encouragement for local residents to seek further education and increase the percentage of adults in Red Creek with bachelor's degrees or higher, which was only half that of New York State as of 2021. This makes the providing of jobs an essential priority in this region, hence ranking the first in our three major factors.

An increase in the region's productional capacities and contribution to the GDP is also an extremely important factor in our decision-making metric. Due to the region the plot of land is found to be relatively isolated from urban centers, there are fewer commercial activities taking place and less money flowing into the economy. Thus, attracting opportunities to generate money flow and expand the local economy will doubtless benefit communities in the vicinity of the project, shaping a better environment for Red Creek and with a sound proposal, even surrounding villages. This benefit will be obvious through improvements in the life quality of nearby residents such that it can bring about an increase in GDP contribution, such as like an increase in per-capita income. As of 2021, the average income for a person living in Red Creek was about $26,000 annually, 80 percent of the average being within Cayuga county and only three-fifths that of New York State. By adding a larger weight to the importance of economic contribution within our metric, we seek to ensure that more money is able to flow into the economy of Red Creek and that the planned use of the land can not only generate profit but also allow surrounding communities to reap the fruits of its success.

Last of our major considerations is the environmental impact of both the construction and long-term sustainment and maintenance of the project under consideration. As the era of climate change continues, the reduction of pollutant creation and emission has become of utmost significance in order to maintain the quality of the environment. This becomes especially important due to the fact that the plot of land under consideration is located just off the East coast of the USA, where there are significantly more pollutants such as particulate matter than can be found elsewhere in the United States. Therefore, this factor being taken into account can also address the environmental side of this problem and provide insurance that there will be no significant damage on the habitat.

Other than these three major factors that we have decided on, though, there is another factor that may arguably be even more important, namely profit. No matter the project, it would be unfeasible to produce a plan that would generate negative gain. Naturally, profit is considered to be a beneficial factor to our group. However, we did not include it in our list of major factors, as the importance of those factors was weighted through their importance in regard to contribution to the community, and profit would not have been able to be included. In our final algorithm and model, we will rank community benefit and profit produced separately, and our final judgment will be based on a score relying on the contribution of both variables.

5.2 Task 1

In the question, we have identified all of the factors that will affect the benefits that can be attained. They are: land, cost upfront, jobs, attraction, cost sustained, earn, environment, and tourism. We then grouped these 8 factors into three main groups: economy, lands, and humans.

5.2.1 Economy (M_{cu}, M_e, M_{cs})

Our team decided to calculate the GDP of the project, and the general income of the project. The GDP can accurately calculate and display the impact of the project on the entire society as a whole, providing a visual number showing how much the different reconstructions can help the society on a financial level. The profit will be included because profit is always important to the project.

Since M_e and M_{cs} are based on difference of money per year, it may not be a constant. For example, for M_{ce} we decided to calculate the GDP and the profit over the length of 10 years. Using the factors M_{cu}, M_{cs}, M_e, and time, we composed the following equation:

$$GDP = M_{cu} + (M_e + M_{cs}) * 10$$

$$profit = (M_e - M_{cs})\times 10$$

$$eco = GDP + profit$$

5.2.2 Lands (land, environment)

Land is a vague topic to consider. Forests are evidently better than plains, as the cost of trees outweighs the cost of crops, but after research we find that it is impossible to calculate how much more value 1 m^2 of forest bears when compared to 1 m^2 of cropland due to too many possible fluctuations in cost (but the range of trees was still higher than that of the crop cost range). Our team decided to view the forest as trees growing on cropland. The difference in the values of the forest and cropland can be seen as the value of the trees in the forest. A pine tree is worth around $30, and there are about 0.02 trees in 1 m^2 (on average). Therefore, 1 m^2 of forest land is worth about $0.9 more than the same area of cropland. One block of land, in our definition, is 290 m^2, so we calculate that each block of land would be worth a total of $210.

Another factor to consider is that growing crops will deplete nutrients in the soil at a certain rate. To describe the depletion, we added a new variable to calculate the land. The depletion of the soil will be calculated as crop land * (1 - depletion) ^{time}.

Considering these two factors, we composed an equation for land

$$Env =$$

$$(l_f - l_{fa})\times 210 + (l_{ca} \times (1 - dp)^{10}$$

5.2.3 Humans (jobs, attraction, tourism)

The factor related to humans is the hardest to measure. Jobs and attraction can be represented in single numbers, but tourism depends on multiple factors and can only be estimated as the number of components is simply immeasurable.

In addition, it is acknowledged that the number of tourists who would arrive would be much larger than the number of jobs or attraction, so we need to find a way to balance the three components to ensure there are enough workers to provide services for the tourists. To balance the three components, we estimated the largest number of tourist possible and the largest number of jobs possible, as shown below:

$$\frac{A_t}{decreased\ A_t} = \frac{tourist\ max}{jobs\ max} = \frac{6000}{15}$$

Using this formula, the decreased tourist can be balanced equally with the jobs and attractions.

Combining the factors of jobs, attraction, and tourism, we get the ultimate equation, shown below:

$$Hum = A_e + A_{mw} + A_t \times \frac{jobs\ max}{tourist\ max}$$

The final benefit cannot be easily adding them up, because the scale of the eco, env and hum is different. From observations along with trial and error, we discovered that the eco should be divided by 10^7, env should be divided by 10^4, and hum should not be scaled. Some more trial and error, we located the good coefficients of each factor. The final benefit equation looks like this:

$$Benefit = \frac{5}{8} \times (0.35 \times eco \times 10^{-7} + 0.6 \times env \times 10^{-4} + 0.7 \times hum)$$

5.3 Task 2

Task 2 requires that we weigh the values of the different choices and decide the most beneficial options among the given ones according to the factors that we determined as the most important. Therefore, extensive research was conducted in order to gather the needed statistics for later comparison. This allowed us to gain a more objective perception on the entire problem as we were able to find the numbers from credible sources, as shown in the references section. The figures were then compiled into a visual graph for easier differentiation and later analysis, as shown below:

	Land	cost upfront	jobs	attraction	ost sustainec	earn	environment	toruism		benefit
Solar Farm	900000 m2 c	161936350	10	0	675510	255600000	2100000 m2	0		79
	1200000 m2 forest						0 m2 forest			
	540000 m2 wetland						540000 m2 wetland			
Lettuce	900000 m2 c	630000	10	0	62533605	107806400	2100000 m2	0		52
	1164950 m2 forest						0 m2 forest			
	540000 m2 wetland						540000 m2 wetland			

As portrayed in the table, the final result that we got that was the best suited according to the land's landscape, resource availability, and climate, were a solar array and a crop farm growing lettuce.

5.4 Task 3

Task 3 states that a new semiconductor factory would be built North of Syracuse, employing a large number of workers. Our team believes that the factory would affect our piece of land in Red Creek in the following ways:

1. Increasing electricity bills: since 9,000 people will be working in the large factory producing semiconductors, a great amount of energy will be required. The demand of electricity will therefore be increased. Since we have no information of the local energy production, we can only estimate the rising of the electricity cost. The estimated value is 1.1, representing the electricity cost inflates by 10%.

2. Decreasing job attraction: as the large fabrication facility will create 40,000 jobs, more people will be attracted to that factory as opposed to the jobs offered during the reconstruction. Therefore, we might face problems in hiring workers if our reconstruction of

the land requires many employees. The introduction of the semiconductor facilitation facility will introduce a large range of new jobs for people living in the region to apply for, along with a salary significantly higher than the regional average. This will decrease the temptation that other jobs, including those introduced by our construction project, have, meaning we must increase salaries or face the risk of becoming understaffed. Therefore, the importance of manpower and jobs within our planned initiative must be decreased in order to lower expenditure and the reliance on manpower which may not be available if migration attraction is not high enough. Again, these numbers need to be estimated, and therefore we believe that 5 workers will stay.

3. Increasing population: the large factory might be able to attract migrant workers and therefore increase the population of the region around it. The increasing population would be beneficial to most industries near Syracuse as it would see to an increase of available people for service. There are 40000 additional jobs, which means that at least 40000 people will come to Clay, NY. Since the distance between the project and Clay, is quite far, the impact of this sudden growth in population would not be significant. We can reasonably assume that 1% of the people will need to come to Red Creek and its nearby area. 400 people will likely come to Red Creek. There will be a 6% increase in the total population, therefore the benefit of the project should also increase by 6%.

To reevaluate our metric, we devised a plan for the corresponding situations. As for increasing electricity bills, we can modify our model by relying on renewable energy sources more, such as they can supply great amounts of energy. The earnings of building solar panel arrays might therefore be increased. When it comes to decreasing job attraction, it may not be beneficial to any way of using the land, since all of them require employees. People will be attracted to the semiconductor factory and we would have to pay more wage to workers. At last, population growth would be beneficial to most options of using the land, since a population growth would result in a growth of consumption of resources.

The benefit of the solar farm increases 14, reaching 93. The benefit of the lettuce farm increases by 1, which is 53.

5.5 Task 4

When our team was considering the task, we found that there were a great number of differences between the piece of land described in the problem we were assigned and the environments we were familiar with, our model may be not suitable in this situation. To test the accuracy, our team compared the three main factors affecting the model: environmental effects, human effects and economic effects (GDP).

The region our team is most familiar with is the CBD of a large city, namely Shanghai. As a result, the population under consideration would be significantly larger.

One effect of the large population and dense urban center is developed industry and well equipped facilities. Therefore, the air would be greatly polluted when compared to the plot of land considered within the first few parts of this paper and some environmental-friendly structures may rank higher in our list of priorities as excess pollution cannot be afforded by Shanghai's environment. Although Shanghai is seen as a famous tourism center, tourism might not be the best solution precisely due to that fact; large cities have enormous amusement parks such as Disneyland or water parks (as is the case for Shanghai), decreasing the appeal of smaller amusement centers we would consider building when compared to more famous parks. If we really were to design a more luxurious entertainment facility similar to Disneyland, the cost of the construction and design would be significantly more than all of the other options and too costly to be carried out. Another reasonable choice is to build an area for residence within the 3.1km^2 region. Since large cities have densely populated CBDs and high rent, a series of buildings or houses for living could contribute greatly to the GDP of the city. Places of residence, unlike factories, would not have much effect on the environment either relatively. A third option would be to build a solar panel array and/or farm. Nevertheless, these structures should not be near urban areas where land is scarce. Instead, they should be built in rural regions as a supply for energy. Using renewable energy such as wind power and solar power would be environment friendly. Considering the environment we are familiar with, other options such as building factories or building infrastructure may not be helpful, since large cities have well developed industry, and the already present pollution cannot afford for another increase from factories and such.

Despite the fact that the paragraph above gives a brief explanation of manipulating the model, in order to have accurate data we need to input the data into our model and calculate the effects the different constructions will have on the environment, on the society and on economics precisely. Our model may be generalizable to most countries and regions since the data can be inputted directly. However, for some special regions or undeveloped countries, our model may not be completely accurate. For example, if our model calculates that the best method is to build solar arrays in a place that does not receive much sunlight, it would not be the best solution in reality.

In conclusion, after actual comparison to see whether or not our model could be applicable to our familiar place—Shanghai—we saw that the gap between the 2 compared regions were too different to allow for a proper analysis. However, we are sure that it can be applied to most regions with just inputting the corresponding data, but it would definitely generate unwanted results if the criteria simply did not match those of the model.

V. Strengths

Strength 1: The model that we created for this problem is easy to use. It only requires data for 8 variables in which just needs to be plugged in to get the final plan for land usage, so there will be no need to go to complicated measures to collect excess amounts of information to ensure that the main factors are all taken into consideration. It rids the planning process of superfluous procedures so that the problem can be solved both efficiently and correctly.

Strength 2: Our metric can be easily applied to different regions as long as the statistics fit the criteria of our metric since it is concise. It is not restricted to only a specific kind of region, so this enables the broadening of the range of regions which can be planned so that the area is used to their full extent, and in the long run it can definitely serve as reference for the blueprint for a wide range of areas.

VI. Weaknesses

Weakness 1: During the making of our model, we used estimations, or averages, during the comparison of value between different options for the given piece of land in Red Creek. However, it was hard to be able to find one exact answer for a value we wanted, say, the cost for one pine tree, since there were fluctuations between the market as well as the quality and height of the tree. Therefore, we took the average cost of the ranges and made estimations that were best, so even though it wasn't the exact number, we were still able to ensure that the model did not stray away from our expected accuracy.

Weakness 2: We did not take into much consideration the social angle during the process of constructing our model. We do see that there will definitely be exceptions in the behavior of tourists and their satisfaction of our reconstruction, which may have an impact on our income and therefore ultimate benefit. However, compared to the other main factors that we identified in the previous sections of the paper, the social factor does not yield so big an impact, so the absence of it would not generate an unsatisfactory result.

VII. Recommendation Letter

To the community leaders and business planners:

Greetings! The major factors we have chosen to take into consideration include GDP, jobs provided, and environmental impact of the recommended proposals. In our model we have decided to weigh the importance of these three factors depending on their relative value to the assigned region, and after comparison we believe the highest emphasis should be placed on the factor of available jobs to increase employment rates and decrease poverty in the region. According to statistical data from the ACS 5-year estimates (US Census Bureau), the proportion of those suffering from poverty in the Red Creek Central School District is approximately 1.3 times that of the surrounding Cayuga County and New York State, so if jobs with stronger expertise in certain regions could be created, it would also serve as encouragement for local residents to seek further education and increase the percentage of adults in Red Creek with bachelor's degrees or higher, which was only half that of New York State as of 2021. This makes the providing of jobs an essential priority in this region, hence ranking the first in our three major factors.

An increase in the region's productional capacities and contribution to the GDP is also an extremely important factor in our decision-making metric. Due to the region the plot of land is situated in being relatively isolated from urban centers, there are few commercial activities and therefore less money that is flowing into the economy. Thus, attracting opportunities to promote money flow and expanding the local economy will undoubtedly benefit communities in the vicinity of the project, shaping a better environment for Red Creek and with a sound proposal, even surrounding villages. This benefit will be obvious through improvements in life quality of nearby residents that an increase in GDP contribution might bring, such as increase in per-capita income. As of 2021, the average income for a person living in Red Creek was about $26,000 annually, 80 percent of the average within Cayuga county and only three-fifths that of New York State. By adding more emphasis to the importance of economic contribution within our metric, we seek to ensure that more money is able to flow into the economy of Red Creek and that the planned use of the land can not only generate profit but also allow surrounding communities to reap the fruits of its success.

Last of our major considerations is the environmental impact of both the construction and long-term sustainment and maintenance of the project under consideration. As the era of climate change continues, the reduction of pollutant creation and emission has become of utmost significance in order to maintain the quality of the environment. This becomes especially important due to the fact that the plot of land under consideration is located just off the East coast of the USA, where there are significantly more pollutants such as particulate matter than can be found elsewhere in the United States.

However, other than these three major factors that we have decided on, there is another factor that may arguably be even more important, namely profit. No matter the project, it would be unfeasible to allow a plan that would generate negative gain. Naturally, profit is considered to be a beneficial factor to our group. However, we did not include it in our list of major factors, as the importance of those factors was weighted through their importance in regard to contribution to the community, and profit would not have been able to be included. In our final algorithm and model, we will rank community benefit and profit produced separately, and our final judgment will be based on a score relying on the contribution of both variables.

Regards,

IMMC Team 23255691

VIII. Conclusion

To conclude, our metric combined attraction, environment and economics into consideration and after trial and error we resulted in a function calculating the weights of each factor, finally achieving the ultimate result: that the economics or GDP of the project bears the highest significance to the general prosperity of the land and beyond, followed by environmental and population factors.

For task 1, we further analyzed each factor by calculating a value determining the effect of our project. Then, we used the data of the three factors to calculate the final benefit value with coefficients that were calculated using a combination of the Cauchy-Schwartz inequality and iteration.

Using the data from task 1, we found out the way of using the land with the highest benefit value: solar array and a crop farm growing lettuce. It best suited the unique landscape, climate and resource availability of the region, and is therefore considered to be the optimal solution by our metric.

As for the effect the semiconductor factory would have upon our reconstruction, we predicted there to be an increase in electricity bills due to a higher demand for energy, a decrease in job attraction of migrant workers towards our land and an increase in the population number of red creek. After calculation, we found that both the benefits of the solar array and the lettuce farm increased.

At last, we compared and generalized our model by switching the location of the land from a small village to the CBD of a large city, Shanghai. In the process, we had to change some constants and more options were taken into consideration as there were definite differences between the two regions.

All in all, our metric is a comprehensible model that can be efficiently used to calculate the best way of using land as long as the statistics fit our criteria. However, our model does not include special scenarios such as the behavior of tourists, and therefore may be inaccurate in some cases.

IX. References

"1 Mega-Watt Solar Kits | SunWatts." *Sunwatts.com*, sunwatts.com/1-mega-watt-solar-kits/. Accessed 10 Mar. 2023.

"CalTopo - Backcountry Mapping Evolved." *Caltopo.com*, caltopo.com/map.html#ll=43.22923. Accessed 10 Mar. 2023.

Cambridge Dictionary. "Up-Front Fee." *@CambridgeWords*, 8 Mar. 2023, dictionary.cambridge.org/dictionary/english/up-front-fee#:~:text=an%20amount%20of%20money%20paid. Accessed 13 Mar. 2023.

"Census Profile: Red Creek Central School District, NY." *Census Reporter*, censusreporter.org/profiles/97000US3624210-red-creek-central-school-district-ny/. Accessed 10 Mar. 2023.

Conner, Timothy. "How to Sell Solar Energy – Alternative Energy." *Altenergyoptions.com*, altenergyoptions.com/how-to-sell-solar-energy/. Accessed 10 Mar. 2023.

"Crops | National Geographic Society." *Education.nationalgeographic.org*, education.nationalgeographic.org/resource/crop/.

"Definition: Sustained Questioned Costs from 10 USC § 2313a(D)(2) | LII / Legal Information Institute." *Www.law.cornell.edu*, www.law.cornell.edu/definitions/uscode.php?width=840&height=800&iframe=true&def_id=10-USC-1260486711-182411289&term_occur=999&term_src=title:10:subtitle:A:part:IV:chapter:137:section:2313a#:~:text=(2)%20The%20term%20%E2%80%9Csustained. Accessed 13 Mar. 2023.

Freitas, Taylor. "The Hidden Costs of Solar Panels: Price of Solar Panel Maintenance and Repairs." *CNET*, www.cnet.com/home/energy-and-utilities/what-are-the-hidden-costs-of-solar-panels/.

Gonzalez, Gerardo. "How Much Does It Cost to Grow an Acre of Lettuce?" *Sweetish Hill*, 7 Aug. 2022, sweetishhill.com/how-much-does-it-cost-to-grow-an-acre-of-lettuce/. Accessed 10 Mar. 2023.

"How Much Energy Do Solar Panels Produce in a Year? Understanding Average Solar Panel Output." *Www.rocketsolar.com*, www.rocketsolar.com/learn/solar-basics/how-much-energy-do-solar-panels-produce.

Kilgore, Georgette. "Tree Value Calculator: 23 Species Ranked by Value (Updated 2023 Prices)." *8 Billion Trees: Carbon Offset Projects & Ecological Footprint Calculators*, 26 Jan. 2023, 8billiontrees.com/trees/tree-value-calculator/#:~:text=Large%20pine%20trees%20on%20average. Accessed 10 Mar. 2023.

Krymowski, Jaclyn. "The True Financial Cost of Modern Farming." *AGDAILY*, 8 Oct. 2020, www.agdaily.com/insights/true-financial-cost-modern-farming/.

Marsh, Jacob. "Types of Solar Panels: What You Need to Know | EnergySage." *Solar News*, 7 Feb. 2021, news.energysage.com/types-of-solar-panels/.

Metaye, Romain. "Monocrystalline Solar Panel — Everything You Need to Know." *Climatebiz*, 27 Feb. 2021, climatebiz.com/monocrystalline-solar-panel/.

"Number of Trees per Km^2." *Our World in Data*, ourworldindata.org/grapher/number-of-trees-per-km. Accessed 10 Mar. 2023.

"Page Analysis» Selina Wamucii." *Selina Wamucii*, www.selinawamucii.com/insights/prices/united-states-of-america/lettuce/.

Tasgal, Peter. "The Economics of Vertical & Greenhouse Farming Are Getting Competitive." *AFN*, 3 Apr. 2019, agfundernews.com/the-economics-of-local-vertical-and-greenhouse-farming-are-getting-competitive.

Winner, George. *New York NYS Legislative Commission on Rural Resources * Senator Land Use Tools Counties,Cities, Towns, and Villages a 2008 Survey of Land Use Planning & Regulations in NYS.*

X. Appendix

```
def economics(cost_upfront, earn, cost_sustained):
    #Amount fo money used for the project to sustain itself
    GDP = cost_upfront + (cost_sustained + earn) * 10
    return int(GDP) + (cost_sustained + earn) * 10


def lands(land, environment):
    #Assume 1 block of crop land is priced 1, forest is prices as trees which is 210
```

```
    land_after = environment[0]

    depletion = environment[1]

    f2c = land['forest'] - land_after['forest']

    f2c_minus = 210

    return -((f2c*f2c_minus)  + (land_after['crop'] * (depletion)**10))


def humans(jobs, attraction, tourism):

    #assume jobs_max ==> 15, tourism_max ==> 6000

    tourism_max = 6000

    jobs_max = 15

    tourism_co = jobs_max / tourism_max

    return jobs + attraction + tourism * tourism_co


#local coefficient = 1

#vacation coefficient = 1/3

#tourism = local * local coefficient + vacation * vacation coefficient = 36000 * 0.1 * 1 +
(265.5 * 10**6 * 10**-4) * 1/3 = 3600 * 1 + 26550 * 1/3 = 12450 people

#tourism = local + vacation

#local = local people * cost_upfront * time * 10        time = 1

#vacation = vacation people * cost_upfront * importance * time         importance = 0.01, time
= 0.01

#local = 36000 * 2700000 * 1


def benefit(land, cost_upfront, jobs, attraction, cost_sustained, earn, environment, tourism):

    #land should be in a dict: describing each type of block and how much in each type |
{crops: x, forest: y, wetland: z}

    '''visualize'''

    #cost_upfornt should be a number: describing the cost of building the land | $money
```

'''reasearch'''

#jobs should be a dict: describing how much different jobs there are in the land | human

'''research'''

#attraction should be a number: describing how much people will come to this farm | human

'''unknown'''

#earn should be a number: it describes how much money the land use will make per year | $money

'''research'''

#cost_sustained should be a number: describing how much money should be used to maintain the land and give the salary per year | $money

'''research'''

#environment should be a dict: describing each type of block and how much in each type after stuff ({crops: (x + a), forest: y - a, wetland: z}, q)

'''research'''

#tourism should be a number: describing how much people will vist the land per year | human

'''estimate'''

```
eco = economics(cost_upfront, cost_sustained, earn)
env = lands(land, environment)
hum = humans(jobs, attraction, tourism)
eco = int(eco / (10**7))
env = int(env / (10**4))
hum = int(hum)
return int(5/8 * (0.35 * eco + 0.6 * env + 0.7 * hum))
```

```
land = {"crop": 2961, "forest": 5065, "wetland": 2299}
#1 block = 230 m^2
salary = 67521
```

```
# Solar Farm 163 MW solar panel

consumption =  7.2 # 60000 households in the vicinity
production = 28.3 #568*5/25/4
requirement = consumption/production
print(f"Requirement coficient: {requirement}")
cost_upfront = 163000000
earn = 568 * 450000  #38699460 #16000 * 1000 / 40000 * 163 / 568 * 450000
cost_sustained = 300 + 10 * salary #300 + 5 * 67,521
environment = ({"crop": 2961 + 5065, "forest": 5065 - 5065, "wetland": 2299}, 1)
jobs = 10
attraction = 0
tourism = 0

print(f"Benefit of Solar Farm: {benefit(land, cost_upfront, jobs, attraction, cost_sustained,
earn, environment, tourism)}")

#Crop farm Lettuce farming for 2.3 km^2 (569 acres)

consumption = 136000
prduction = 568 * 12000 * (365 / 20)
requirement = consumption/production
print(f"Requirement coficient: {requirement}")

cost_upfront = 630000 #310,000 + 300,000 + 20,000
earn = 10400 * 568 * (365 / 20) #$2.09 * 568 * 12000 * (365 / 30) / #$10400 * 568 * (365 /
30)
```

```
cost_sustained = 568 * 9000 * (365 / 20) + 10 * salary #568 * 9000 * (365 / 30) + 5 * 67521

environment = ({"crop": 2961 + 5065, "forest": 5065 - 5065, "wetland": 2299}, 0.1)

jobs = 10

attraction = 0

tourism = 0



print(f"Benefit of lettuce farm: {benefit(land, cost_upfront, jobs, attraction, cost_sustained, earn, environment, tourism)}")


#If a new factory is built at east of the land:

#Affect 1. electricity cost


# Solar Farm 163 MW solar panel


consumption =  7.2 + 614 # 60000 households in the vicinity

production = 28.3 #568*5/25/4

requirement = consumption/production


print(f"Requirement coficient: {requirement}")


cost_upfront = 163000000

earn = (568 * 450000) * 1.1 #38699460 #16000 * 1000 / 40000 * 163 / 568 * 450000 1.1 is unimportant, an estimate price due to the demand of electricity will increase but do not really need this

cost_sustained = 300 + 5 * salary #300 + 5 * 67,521

environment = ({"crop": 2961 + 5065, "forest": 5065 - 5065, "wetland": 2299}, 1)

jobs = 5 #cut half estimation
```

```
attraction = 0

tourism = 0

print(f"Benefit of Solar Farm with factory: {int(benefit(land, cost_upfront, jobs, attraction, cost_sustained, earn, environment, tourism) * 1.06)}") # (9000 * 0.5 * 0.1 + 6000) / 6000


#Crop farm Lettuce farming for 2.3 km^2 (569 acres)


consumption = 136000

production = 568 * 12000 * (365 / 20)

requirement = consumption/production

print(f"Requirement coficient: {requirement}")


cost_upfront = 630000 #310,000 + 300,000 + 20,000

earn = 10400 * 568 * (365 / 20) #$2.09 * 568 * 12000 * (365 / 30) / #$10400 * 568 * (365 / 30)

cost_sustained = 568 * 9000 * (365 / 20) + 5 * salary #568 * 9000 * (365 / 30) + 5 * 67521

environment = ({"crop": 2961 + 5065, "forest": 5065 - 5065, "wetland": 2299}, 0.1)

jobs = 5

attraction = 0

tourism = 0


print(f"Benefit of Lettuce with factory: {int(benefit(land, cost_upfront, jobs, attraction, cost_sustained, earn, environment, tourism) * 1.06)}")
```

Finding parameters of benefit

```
import random


def benefit():
```

```
    eco = random.randint(1000000000, 3000000000)
    env = random.randint(900000, 1000000)
    hum = random.randint(0, 20)



    return int(5/8 * (0.7 * eco / 10**7 - 0.6 * env / 10**4 + 0.7 * hum))
num = []
for i in range(1000):
    num.append(benefit())

print((sum(num) / len(num)))
print(min(num), max(num))
```